The microbial ecology of permafrost

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Abstract | Permafrost constitutes a major portion of the terrestrial cryosphere of the Earth and is a unique ecological niche for cold-adapted microorganisms. There is a relatively high microbial diversity in permafrost, although there is some variation in community composition across different permafrost features and between sites. Some microorganisms are even active at subzero temperatures in permafrost. An emerging concern is the impact of climate change and the possibility of subsequent permafrost thaw promoting microbial activity in permafrost, resulting in increased potential for greenhouse-gas emissions. This Review describes new data on the microbial ecology of permafrost and provides a platform for understanding microbial life strategies in frozen soil as well as the impact of climate change on permafrost microorganisms and their functional roles.

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An estimated 25% of the land area on Earth is underlain by permafrost. The classic definition of permafrost is soil that has remained frozen (that is, below 0 °C) for at least 2 consecutive years. However, some ancient permafrost has remained frozen during several periods of glaciation. In fact, permafrost can be up to 1–3 million years old in the Arctic and even older in Antarctica. Depending on the location and climatic conditions at the time, the land surface that is now permafrost was not initially frozen and may have been covered with either vegetation or water. Some of the plant and animal material and microbial biomass that accumulated during the CO_2-rich atmosphere and warm periods in the history of the Earth became preserved in the permafrost during cooling and glaciation events. Although the plants and animals that became trapped during this transition met their demise, current evidence suggests that many of the microorganisms were able to adapt and even remain active. Recent research suggests that permafrost regions will be affected by increases in global temperatures and subsequent permafrost thaw will probably not only result in increased activation of indigenous permafrost microorganisms but will also unlock considerable amounts of carbon and nutrients that can be used for their accelerated growth. In this Review, we summarize our current knowledge of the microbial ecology of permafrost.

Permafrost regions around the globe

Permafrost is a structurally heterogeneous habitat, and there are different types of permafrost in different geographical locations (FIG. 1). Moreover, there are differences within locations with respect to permafrost structure, organic-matter content and age. Permafrost is traditionally classified as either continuous or discontinuous and is characterized by the amount and type of ice that it contains[1]. In general terms, permafrost is different from all other habitats and contains features that are different from unfrozen soil, sediment and rock. Permafrost is overlain by a so-called 'active layer' that has been exposed to seasonal freeze–thaw cycles. The active layer can vary in thickness from a few centimetres to several metres. Owing to global warming, the depth of the active layer has been shown to be increasing in many locations over the past few decades, resulting in a subsequent decline in the underlying permafrost[1–4].

During glacial periods, large regions of the North American and Scandinavian Arctic were covered with ice, and remnants of this ice remain in Greenland and the northeastern Canadian Arctic. By contrast, large regions of the permafrost areas in Asia and Alaska were not recently glaciated but are frozen nonetheless. These include perennially frozen and loess-like (that is, wind-blown) depositions called Yedoma that comprise a specific type of permafrost formation that is mainly found in Northeast Siberia and is rich in organic matter (that is, approximately 2% carbon) with an ice content of 50–90% that can be hundreds of metres thick[4]. Peat deposits are also common in the southern part of the Arctic permafrost region. Substantial variations have been observed in permafrost temperatures, ranging from −2 °C to −10 °C in the Arctic, with some sites below −15 °C (REF. 2).

Although only 0.36% (49,800 km^2) of the Antarctic region is ice-free, permafrost is present beneath almost all of this area[3]. However, there are large differences in water availability across the continent; for example, the Ross Sea region has a strong moisture gradient, in

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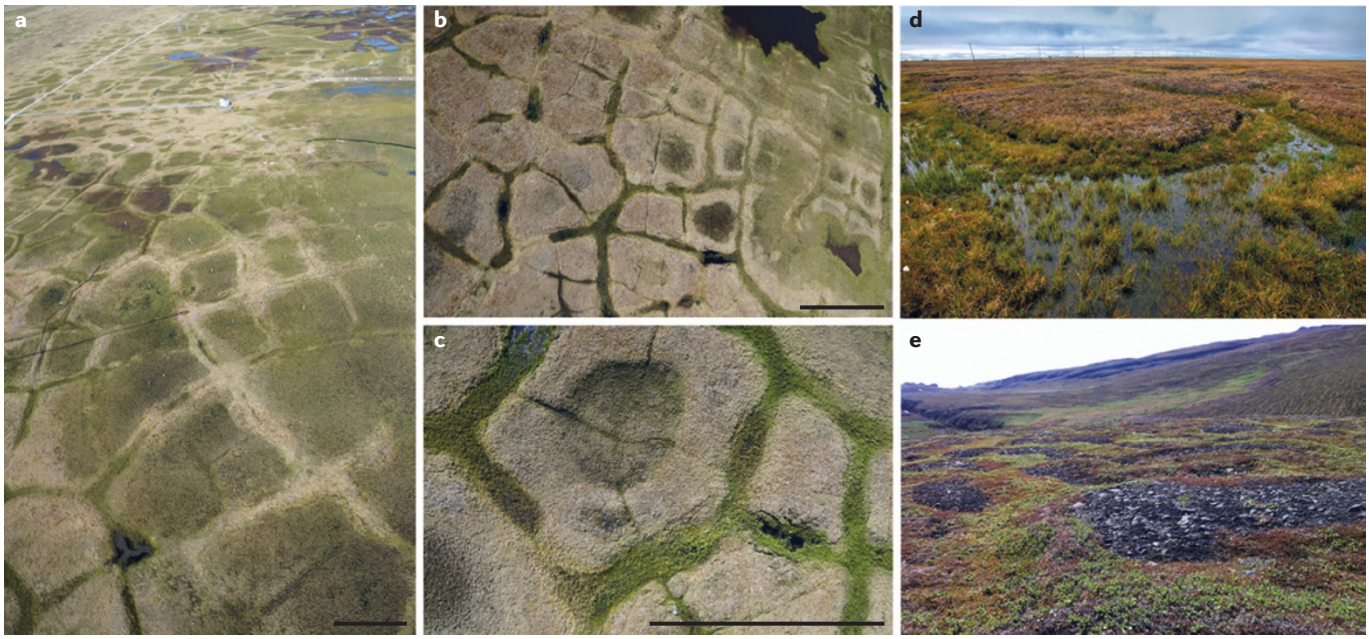

Figure 1 | **Changes in landscape caused by permafrost thaw and cryoperturbation.** **a** | The terrestrial landscape of the Barrow Environmental Observatory (BEO) at the North Slope of Alaska contains a network of polygonal features owing to ice-wedge formation. **b** | The image shows different polygon types: high-centred polygons with raised centres to the left, transitioning to low-centred polygons to the right and a polygonal pond at the top. **c** | The image shows a low-centred polygon with a depressed and wet centre and highly vegetated troughs. **d** | The image shows a flat-centred polygon in transition between high and low-centred polygons. **e** | The image shows unvegetated frost boils (approximately 1.5–2 m in diameter) at Disko Island, Greenland. Scale bars in parts **a**–**c** represent 20 m distances. Images in parts **a**–**c** taken using a kite with attached camera, courtesy of Baptiste Dafflon (Lawrence Berkeley National Laboratory, University of California, USA). Image in part **d** courtesy of Berkeley lab — Roy Kaltschmidt, photographer.

Yedoma
A permafrost type that is mainly found in Northeast Siberia. It is characterized by high levels of organic material and up to 90% ice content. Carbon that is trapped in this permafrost is suggested to be susceptible to microbial decomposition following thaw, resulting in the potential release of large quantities of greenhouse gases into the atmosphere.

Cryopegs
Permafrost zones that have high dissolved-solids contents (for example, a high salt concentration in the pore water), thus depressing the freezing point of water.

Talik
A layer of unfrozen ground that can occur in permafrost zones underlying thermokarst lakes and rivers.

Frost mounds
Hummocks or knolls that are produced by freezing combined with groundwater formation through soil in a permafrost region. Frost mounds contain a core of ice that is covered by a thin soil layer.

which soils contain ice-cemented permafrost, whereas most inland areas receive less than 50 mm precipitation per year and the soils are dry-frozen[5]. The permafrost in Antarctica can be very thick — up to 680 m in depth[6] — and temperatures vary between −1.8 °C and −24 °C. Antarctic soils are coarse-textured and the soil usually lacks structure, except where high salt concentrations make the soil material firmly cohesive[7]. Antarctic active layers show a thickening trend (1 cm per year), with similar rates to those that are found in the Arctic, where air warming occurs[8].

Temperature changes in permafrost ecosystems result in the formation of several types of landscape features. Although the active layer undergoes seasonal freeze–thaw cycles, the ground below must remain frozen for a minimum of 2 years to be defined as permafrost. However, some permafrost contains unfrozen layers, known as cryopegs, in which the freezing point has been depressed owing to high concentrations of dissolved solids in the pore water[9]. As temperatures are getting warmer, upper permafrost layers are increasingly starting to thaw in the Arctic, which results in an increase in the thickness of the active layer[10]. Depending on weather and soil conditions, the active layer can increase to a thickness that does not completely refreeze in winter, forming a talik[11]. During the winter months, thermal contraction cracks can form in the permafrost; these are consecutively filled by snowmelt water in spring and summer periods. Refreezing of the water inside these cracks in winter months forms ice wedges. Expansion within an ice wedge affects the overlying landscape by separating it into high-centred or low-centred polygons (FIG. 1a–d). Ice wedges are present in both Antarctic and Arctic permafrost, but they are usually small in Antarctica — not more than 0.6 m in width and 1 m in depth[12] — whereas, in the Arctic, they can vary from 0.002 m to 4 m in width and are several meters in depth[13]. Frost mounds are also much less widespread in Antarctica than in the Arctic; this may be due to the more severe summer season or the drier conditions in Antarctica[14,15]. When thawing of ice-rich permafrost or massive ground ice occurs, the ground subsides and the surface is inundated, forming collapse-scar bogs and thermokarst lakes (also known as thaw lakes); this is particularly evident in low-drainage soils. Frost boils, or non-sorted sparsely vegetated soil circles, are another feature that can result from cryoperturbation in areas that have a thin organic active layer and large amounts of snow, and they form owing to differences in the expansion of water during freezing[16] (FIG. 1e). In subzero conditions, the differences between above-ground and below-ground temperatures can cause the migration of soil and water from layers above the permafrost to the surface, followed by freezing, which results in considerable mixing of the soil during frost-boil formation. As a result, carbon and nitrogen are unequally distributed in frost boils and are concentrated in the depressions between the frost-boil formations[17].

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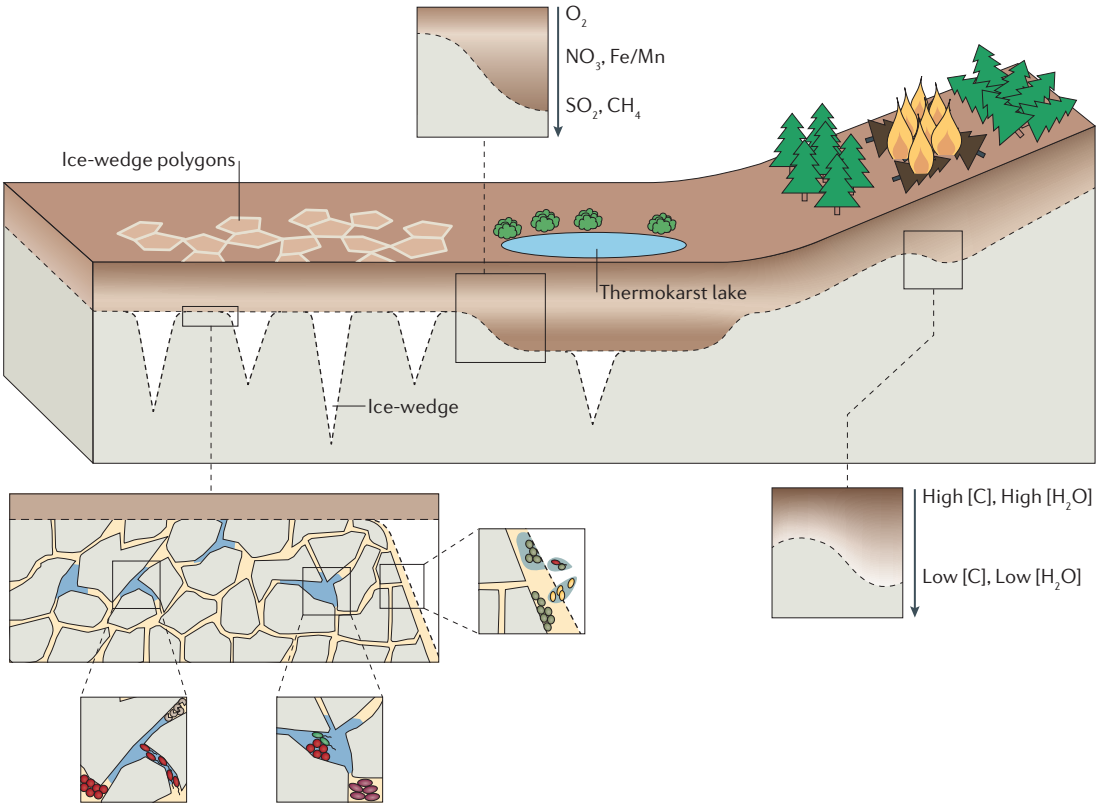

Figure 2 | **Permafrost thaw features at lowland and highland elevations.** The permafrost is overlain by a seasonally thawed active layer (brown). The bold dashed line indicates the surface of the permafrost table. The lowland is characterized by polygons that are separated by ice wedges (white) in the permafrost layer (grey). A thermokarst lake is indicated by a deepening of the active layer and pooling of thaw water at the surface. At high elevations, permafrost thaw results in drainage of the soil moisture and subsidence, which leads to the 'drunken tree' phenomenon. In these locations, permafrost thaw can also originate from the heat that is generated by wildfires. The upper panel and lower right-hand panel indicate differences in redox chemistry, soil and moisture with depth. The lower left-hand panels show close-ups of individual soil microaggregates (brown, active layer; grey, permafrost) and microcolonies of bacterial or archaeal cells in the pores containing free water — that is, brine veins. Figure is not drawn to scale.

Thermokarst lakes
(Also known as thaw lakes). Shallow, freshwater bodies that are formed by the collapse of the underlying permafrost and the accumulation of meltwater as permafrost thaws in depression areas.

Frost boils
Sparsely vegetated circular features (with a diameter of 0.5–3 m) on the land surface; they are formed by the uplifting of mud that is formed by thawing of the below-ground permafrost.

Owing to climate change, there have also been more wildfires in the Arctic. Recent research has shown that fire results in intense heating of the soil surface and the consecutive thaw of near-surface permafrost[18]. Fire is thus a major factor to take into account when considering the long-term dynamics of permafrost stability in boreal ecosystems[19]. Particularly in upland forests that have mineral soils and considerable slope, the fire-mediated rapid permafrost thaw and the subsequent drainage of permafrost water has been shown to create drier soils and to alter biogeochemical cycling processes[19,20].

Permafrost as a microbial habitat

Permafrost is a unique habitat for microbial life on Earth and is considered to be a vast repository of ancient viable cells[21]. Considerable excitement was recently raised by the regeneration of ancient giant DNA viruses[22] and moss species[23] from permafrost. These examples highlight the recognized potential for the preservation of microbial life (that is, Bacteria, Archaea and Fungi) in a dormant state in permafrost[21]. What distinguishes permafrost from other low-temperature environments, such as sea ice, glaciers and deep oceans, is its structural heterogeneity, as it has both horizontal and vertical differences in soil or sediment texture, ice content and organic-matter content (FIG. 2). Probably the closest correlation is to deep marine sediments, which are cold and vertically stratified with respect to oxygen content, carbon content and texture. It is intriguing to think of the potential for life in permafrost from an evolutionary perspective. According to the 'Snowball Earth' hypothesis, the Earth has been completely frozen one or more times as recently as 650 million years ago[24], so microorganisms would have had to be able to adapt to subzero temperatures to continue to live on our planet. Permafrost is also considered to be an example of what life could be like on cold planets, including Mars (BOX 1).

In general, there is considerable variability in the diversity and composition of microorganisms that are found in permafrost, which could be due to differences in specific conditions within the soil and ice. For example, cryopegs that have a high salt content can depress the freezing point and function as a habitat for microbial life in permafrost[25,26]. Microbial cells can also be concentrated at the top of the permafrost table, which represents a physical boundary between the active layer

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Box 1 | **Permafrost as a model of an extraterrestrial biome**

Permafrost is thought to be a common universal feature beyond our own planet. For example, six of the planets in our own solar system, and their moons and comets, are cryogenic in nature. On Mars, permafrost is predicted to exist in most of the subsurface of the planet, where the expected thickness varies between the poles (3–7 km) and the equator (1– 3 km)[112]. In recent years, high-resolution imaging confirmed geomorphological features that are characteristic of active-layer processes that were likely to have formed in the Martian geological past under different climate conditions[113]. Among these features, there are examples that are strikingly similar to terrestrial ice-wedge polygons on Earth. The probable presence of liquid water in the Martian subsurface as brine veins between ice crystals or in permafrost suggests the possibility that there are living microorganisms in this particular niche. These microorganisms would be expected to exhibit strong adaptive characteristics for survival at high osmotic pressure and low temperatures, like their counterparts on Earth. This possibility has spurred the study of permafrost as an analogue to life on other planets. In particular, the detection of the presence of viable cells within permafrost on our planet provides impetus to search for life in other cryogenic bodies and to understand possible life strategies for biological entities in cold environments in general. In addition, the study of the characteristics of microorganisms in permafrost is providing specific targets for astrobiology research and for the detection of signs of life in other celestial bodies.

and the permafrost owing to the infiltration of melted water through the upper soil layers[27]. Ice wedges within the permafrost can also support microbial life in liquid brine veins and have been postulated to have relatively high cell abundances, as the microorganisms are concentrated from the water that percolates through the soil[28]. In frost boils, the microbial composition is heterogenous, depending on the concentration of carbon and nitrogen and the amount of drainage[17,29].

Limits for microbial life in permafrost

Microorganisms in permafrost have traditionally been generally classified as psychrophiles, or more recently as 'cryophiles', which are capable of growth and reproduction at low temperatures, typically ranging from −17° C to +10 °C (REF. 30). The use of minimum and maximum growth temperatures to define microbial cold-adaptations is a subject of ongoing debate[31,32]. So far, the lowest temperature limit that has been reported for bacterial activity in permafrost soil is −17 °C (REF. 33). Temperature and water availability are thought to be the most important determinants of biomass levels, bacterial diversity and community structure in low-temperature environments in general[34–36]. Microbial survival in permafrost involves additional challenges, including low thermal energy[37] and high viscosity[38]. In cold temperatures, proteins are less flexible and are prone to cold denaturation, cell membranes often lose their fluidity, which affects nutrient transport[39], and nucleic acids are more stable, resulting in inhibition of replication, transcription and translation[40].

Microorganisms have evolved to have several strategies for survival at subzero temperatures (BOX 2). These include entering a dormant state with low metabolic activity and producing specialized proteins or other molecules[41] that enable the cells to survive[40]. Cold-adapted microorganisms generally need special adaptations for their cellular components to remain fluid and functional at subzero temperatures — for example, shifting of their phospholipid metabolism to maintain membrane fluidity and regulation of branched-to-saturated fatty-acid production in their cell membranes[42–44]. In addition, as low temperature slows metabolism and rates of nutrient uptake, cryophiles often generate carbon and energy reserves in the forms of glycogen, polyhydroxyalkanoates, polyphosphates, triglycerides and wax esters[45].

Microbial diversity and community composition

Although one might assume that permafrost is an inhospitable environment, it in fact hosts a large diversity of microbial life (FIG. 3). Data based on high-throughput sequencing of 16S rRNA genes is becoming increasingly available for a range of permafrost and active-layer samples. The resulting information about the composition of the microbial community reflects the unique and extreme conditions of the permafrost environment. Although some early studies suggested that the microbial diversity in permafrost was comparable to that in the active layer[21], some of our recent sequence data from the Arctic showed that the microbial diversity in permafrost is lower than that in the surface active-layer soils at the studied locations[20]. This is probably due to conditions that are highly selective for cells that can withstand and survive under subzero conditions for prolonged periods of time.

One of the main challenges of studying microorganisms in permafrost using culture-independent methods is the low biomass. For example, Yergeau *et al.*[46] reported 20-fold fewer cells in permafrost than in the active layer. However, the bacterial abundance in different permafrost types can vary considerably, and counts range from 10^5–10^8 cells per gram of dry mass[21]. The isolation of nucleic acids from permafrost is also hampered by the high ice content. However, amplification of 16S rRNA genes, combined with high-throughput sequencing approaches, has enabled the microbial composition to be determined in several permafrost samples[20,46–49]. It is important to remember that the low-temperature conditions in permafrost are also ideal for the preservation of DNA and, therefore, not all of the sequences that are detected may represent active, or even viable, cells. As in other comparisons of culture-based and non-culture-based approaches, the microbial representation in 16S rRNA gene data sets varies from that obtained from cultivation. For example, in the Canadian High Arctic, a predominance of spore-forming Firmicutes strains were isolated from permafrost, although they had a relatively low representation in 16S rRNA gene-clone libraries, which were instead dominated by Actinobacteria and Proteobacteria[50].

The 16S rRNA gene-sequence data have shown that both the Arctic and Antarctic permafrost contain microbial phyla from different functional guilds, many of which are anaerobic, including acetoclastic methanogens, hydrogenotrophic methanogens, sulphate reducers, Fe(III) reducers and denitrifiers[20,21,51–53]. The bacterial diversity in permafrost is generally higher than the diversity of archaea or fungi. Bacterial phyla that commonly occur in 16S rRNA gene data sets from Arctic permafrost include Proteobacteria, Firmicutes, Chloroflexi, Acidobacteria,

Cryogenic
A term used to describe the material state of being at very low temperatures. In biology, the term relates to organisms requiring low temperatures or the effects of low temperatures on organisms.

Brine veins
(Also known as liquid veins). Lines of liquid water that have high salt content within ice; they can transport soluble and insoluble particles under otherwise freezing conditions.

Acetoclastic methanogens
Archaea that produce methane using acetate as a carbon source; they are responsible for approximately two-thirds of the biogenic methane that is produced annually on Earth.

Hydrogenotrophic methanogens
Archaea that produce methane using H_2, CO_2 and sometimes formate as a carbon source.

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Box 2 | **Adaptations in a microbial cell for life at low temperatures**

Regulation of membrane fluidity
Microbial cell membranes must be fluid to facilitate the motions of essential membrane proteins. In cold temperatures, the functional state of the cell membrane changes from a liquid-crystalline phase to a gel phase, which results in the loss of membrane integrity. Cold-adaptive microorganisms must modify the lipid composition of their cell membranes to maintain fluidity at low temperatures[29].

Cold-adapted proteins
Cold-adapted proteins can improve their structural mobility at low temperatures. High activity at low temperatures can be achieved by destabilization of the active site, or of the whole protein, which enables the catalytic centre to be flexible at temperatures that freeze molecular motions[115]. Nucleic-acid-binding proteins that can relieve the adverse effects of low temperatures (such as high DNA stability, impaired access to RNA polymerase and an increase in RNA secondary structures) are proposed to have a central role in cold adaptation[29].

Cold-shock proteins
Cold-shock proteins function as RNA chaperones by minimizing the formation of mRNA secondary structures. As a result, translation efficiency and regulation are not compromised, even at suboptimal temperatures[114].

Cold-acclimation proteins
Cold-acclimation proteins are permanently synthesized during steady-state growth at low temperatures and are proposed to regulate the transcription and translation of the essential proteins that are required for growth[115]. Although they are assumed to be integral to cellular function at low temperature, their roles are still poorly understood.

Antifreeze and ice-binding proteins
Antifreeze proteins protect the cell by lowering the freezing point of water, either by preventing ice nucleation or by inhibiting the formation of larger ice crystals from water molecules. Similarly, ice-binding proteins bind to ice-crystal lattices and prevent them from recrystallizing, thereby protecting the bacterium from freeze-damage of the cytoplasmic membrane[114].

Osmolytes
Osmolytes regulate cell volume and fluid balance during osmosis[116]. They are present in the solution within or outside a cell. Osmolytes can function as osmoprotectants and are particularly important for survival in hypersaline environments at subzero temperatures[116].

Figure 3 | **Microbial composition of permafrost from different geographical locations.** **a** | Taxonomic distribution of bacteria in different permafrost environments[20,28,46,47,50,53,61,63] based on their 16S rRNA gene classifications. Pie charts represent relative abundances of different phyla in a sample set. Note that the differences in community composition could be a result of differences in sample origin as well as differences in the techniques that were used. All studies characterized a single (that is, non-replicated) permafrost sample, except Mackelprang *et al.*[47], which analysed two cores, and Taş *et al.*[20] and Mondav *et al.*[53], which analysed three replicate permafrost samples per sampling location. **b** | Phylogenetic representation of permafrost collected from two samples at Hess Creek, Alaska[47], on the basis of the average relative abundances of 18S and 16S rRNA genes in metagenomic sequence-reads at the domain level (that is, the level of the Eukarya, Archaea and Bacteria). **c** | Phylogenetic representation of permafrost collected from two samples at Hess Creek, Alaska[47], on the basis of the average relative abundances of 18S and 16S rRNA genes in metagenomic sequence-reads at the phylum level for Bacteria. Within each bacterial phylum, several dominant genera were observed.

Actinobacteria (such as Intrasporangiaceae and Rubrobacteriaceae) and Bacteroidetes, as well as several uncharacterized and novel phyla[20,28,46,48,49,54]. In addition, sequences that represent a wide diversity of methanotrophic bacteria (for example, *Methylomicrobium* spp. and *Methylobacter* spp.) have also been detected[20,51,52].

After a short-term thaw, Actinobacteria — particularly representatives of Coriobacteridae and Rubrobacteraceae — were found to increase in relative abundance in permafrost[47]. Rubrobacteraceae are known to be radiation and dessication tolerant[55], thereby providing another hint that these properties are beneficial for survival in permafrost. Some recent reports have also highlighted the abundance of representatives of uncharacterized or candidate phyla in permafrost. For example, several studies have found that uncharacterized representatives of Chloroflexi are ubiquitious in permafrost soils[20,47,56]. However, there are few cultured representatives of Chloroflexi and none of these has close sequence similarity to the sequences that were found in permafrost. We recently reported another example in which the candidate phylum AD3, which has no cultured representatives, was shown to be highly abundant in the permafrost from an upland boreal forest from Alaska[20]. It had previously been reported that AD3 is abundant in subsurface soils that have low carbon contents[57]. In our study, we found that AD3 sequences were particularly abundant in thawed permafrost soils several years after a wildfire event[20], which supports the hypothesis that members of this phylum are adapted to survive in deeper horizons of carbon-depleted soils.

In Antarctica, the desiccating mineral soils of the McMurdo Dry Valleys have a high bacterial diversity[58], and many species occur at low abundance (that is, at high evenness and richness)[59]. Sequences that correspond to Acidobacteria, Actinobacteria, Bacteroidetes and Gammaproteobacteria were found at higher levels at the permafrost interface than at the soil surface[60]. A permafrost core that was 4.2 m deep from Deception Island in Antarctica was recently found to be dominated by the phyla Proteobacteria, Actinobacteria, Bacteroidetes and Cyanobacteria, and Betaproteobacteria was the most abundant class[61].

Archaeal sequences that were detected in permafrost include those from the Euryarchaeota, the Crenarchaeota and the Thaumarchaeota[20,28,47,48,50,62]. In Siberian and alpine permafrost, most of the archaeal community was composed of methanogenic species[63]. By contrast, the archaeal community in ice wedges had low diversity and mainly comprised ammonia-oxidizing Thaumarchaeota[28].

Permafrost isolates

Much of what is known about microbial metabolism in permafrost has been gleaned from enrichment cultures and the study of pure culture isolates from different permafrost habitats[21]. The genomes of some isolates have been sequenced[64–67], and this has facilitated the analysis of the genomic features that enable them to survive in cold-temperature conditions.

Most permafrost isolates so far are bacteria from the Firmicutes and Actinobacteria phyla, although several

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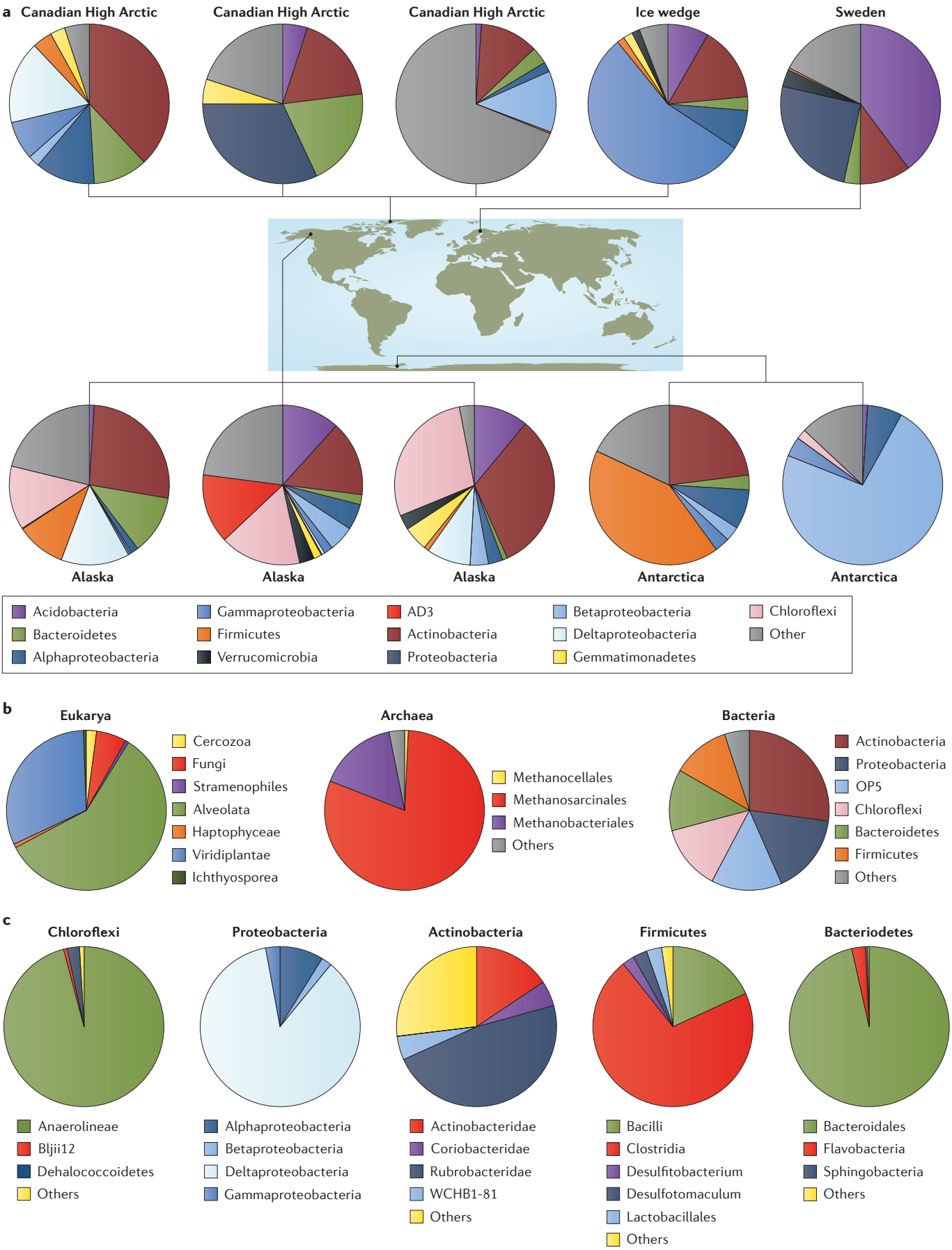

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Bacteroidetes and Proteobacteria have also been isolated (Supplementary Information S1 (table)). Note that several cryophiles have also been obtained from active-layer soils[43,68], but we focus solely on isolates from permafrost. The predominance of isolates from these phyla suggests that they have representatives that are equipped for cold survival and that are also capable of revival and growth in culture media. The ability of many members of the Firmicutes to form spores may be one reason for the high representation of this phylum in culture[50,69]. In addition, some members of the Actinobacteria are known to adapt to stress by entering a dormant, metabolically quiescent but viable state[70]. Furthermore, soil Actinobacteria have a high GC content (65–69%) that may help to reduce DNA damage owing to cold temperatures and high salt concentrations[55]. For example, *Arthrobacter psychrolactophilus* was capable of surviving for 30 days of fluctuating diurnal temperature (from −40 °C to +24 °C) and low pressure (13.3 mbar)[71].

Several studies using different isolation techniques and approaches have shown that viable bacteria are abundant in Siberian permafrost[72,73]. One species, *Exiguobacterium sibiricum* 255–15, was isolated from a 2–3 million-year-old permafrost core at a depth of 43.6 m (REF. 73). The permafrost samples were obtained from the Kolyma–Indigirka lowland, Siberia. The permafrost at this location is considered to be one of the oldest continuously frozen localities on Earth, and it has a stable temperature of −10 °C at a depth of 14 m (REF. 25). The *E. sibiricum* strain was classified as a psychrotroph on the basis of its ability to grow at temperatures ranging from −5 °C to +40 °C. In another study, six bacterial isolates of the genus *Carnobacterium* were obtained from ancient Siberian permafrost (which is 6000–8000 years old) and shown to grow at 0 °C under low pressure (7 mbar) and CO_2-enriched anoxic conditions[74]. This study was conducted as an analogue to life on Mars (BOX 1) and pushed the known boundary conditions for terrestrial life on our planet. Carnobacteria are Gram-positive, and all species that have been identified so far are capable of growth at low temperatures, and can even grow in vacuum-packed meat (hence the name). *Celerinatantimonas yamalensis* — a facultatively anaerobic nitrogen-fixing bacterium — was recently isolated from a permafrost cryopeg from the Yamal Peninsula in Siberia[75]. Although the strain required salts for growth and could grow with up to 12.0% NaCl, it could not grow under freezing conditions.

In addition to bacteria, several archaea have been isolated from permafrost (Supplementary information S1 (table)). The recovery of viable methane-producing strains (known as methanogens), such as *Methanobacterium veterum* and *Methanobacterium arcticum*, suggests that permafrost is a favourable environment for methanogenesis. The minimum growth temperatures for these isolates were 10–45 °C, and they were shown to use CO_2 and H_2 for methane production and growth[62,76,77]. There have also been some studies of fungal isolates from permafrost, including members of the *Geomyces*, *Cladosporium* and *Aspergillus* genera, with *Penicillium* spp. being one of the most frequent isolates[78,79]. Most of these fungi are capable of forming spores — that is, small unicellular conidia — which may enable long-term survival in a dormant state in permafrost[78].

Some permafrost isolates can replicate at temperatures as low as −15 °C (Supplementary information S1 (table)). However, thermodynamic models of microbial growth predict that metabolism can occur at even lower temperatures, close to −40 °C (REF. 80). Studies of some permafrost isolates[43,81–91] suggest that the cells exhibit slow metabolic rates to maintain survival during low-temperature incubations. Laboratory measurements of [^{3}H]leucine and [^{3}H]thymidine incorporation in the Siberian permafrost isolate *Psychrobacter cryohalolentis* showed that the cells were able to synthesize DNA and small amounts of protein (1–16 proteins per day) at −15 °C (REF. 81) (as a point of reference, the average rate of protein synthesis for an *Escherichia coli* cell growing at +30 °C ranges from 10^{-1} to 10^4 molecules per cell in 11–12 hours[92]). In another study, the permafrost isolate *Arthrobacter* sp. 9–2 was shown to incorporate ^{14}C-ethanol during growth at temperatures as low as −17 °C, with an average growth yield of 0.28 g cell C per g C-ethanol (REF. 85). The incorporation of ^{14}C-acetate into lipids was also used to determine the activity of a Siberian permafrost microbial community during incubations between +5 °C and −20 °C (REF. 86). The rate of substrate incorporation decreased with a decrease in temperature, and very slow incorporation was observed at −20 °C. It was proposed that the levels of nutrients became limiting following the formation of diffusion barriers in thin layers of unfrozen water during the incubations[86]. Similar incorporation rates were also measured for some other permafrost isolates[83,84], which suggests that, even with slow metabolic rates, it is possible to compensate for the chromosomal DNA damage that occurs under extreme low-temperature incubation conditions[81].

Genomes of cold-adapted microorganisms have been shown to possess and express genes encoding several features that are required for cold survival, including genes that are required for the maintenance of membrane fluidity, the production of cryoprotectants — including polysaccharides and compatible solutes — and modifications to RNA transcription and protein folding[64,82,93]. Other adaptive mechanisms that have been shown by genome analyses include changes in the compositions of ribosomal proteins and increased expression of protein-disaggregating chaperones, fatty acid desaturases and DEAD-box helicases, which aid in the maintenance of cellular processes despite the thermodynamic constraints[94]. The genome-sequence and gene-expression patterns of the permafrost isolate *E. sibiricum* revealed that the cells undergo considerable adaptation in order to survive and grow at temperatures as low as −2.5 °C (REF. 87). The genes that could be characterized included several that encode translation factors and proteins that are involved in membrane alteration and metabolism. Transcriptome analyses showed that genes in clusters of orthologous groups (COGs) categories for carbohydrate metabolism and DNA replication, recombination and repair, as well as cell division and chromosome partitioning, were differentially expressed[87]. However, many of the genes that

Compatible solutes
(Also known as osmolytes). Small molecules that accumulate in cells to balance the osmotic difference between the inside of the cell and the surroundings of the cell; they help organisms to survive extreme osmotic stress and/or freezing conditions.

DEAD-box helicases
A family of proteins that are involved in the unwinding of RNA. RNA molecules can be mostly single stranded or can adopt specific tertiary structures and are dependent on proteins such as helicases to ensure their correct folding. DEAD-box helicases are also involved in nuclear transcription, pre-mRNA splicing, ribosome biogenesis, nucleocytoplasmic transport, translation, RNA decay and organellar gene expression.

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were expressed had unknown functions. Interestingly, many unknown genes that are likely to be involved in thermal adaptation were differentially upregulated, especially at –2.5 °C. The cells also upregulated transport systems for compatible solutes, such as glycine betain and carnatine, when they were incubated at low temperature. Cryophiles can also contain modifications in their coding genes to compensate for decreased flexibility in protein structure owing to cold temperatures; for example, 56% of the genes in the genome of the permafrost isolate *Psychrobacter arcticus* 273–4 are characterized as cold-adapted, as they encode fewer proline and arginine residues and more lysine residues than corresponding homologues in the UniProt Database[41]. Expression of carbon-metabolizing enzymes and central- and energy-metabolism pathways is generally decreased at low temperatures. However, the increased expression of specific enzymes or isozyme exchange have been proposed as mechanisms to compensate for loss of activity[87,95]; for example, *E. sibiricum*, expresses two different α-amylase isozymes, one at –2.5 °C and the other at 39 °C (REF. 94).

Permafrost metagenomics

Metagenomics has recently been used to study the microbial ecology of permafrost. In addition to avoiding biases owing to cultivation, metagenomics can also reveal which biochemical pathways are present and potentially expressed by different members of the microbial community, including previously unknown and uncharacterized members. Although the inventory of bacteria and archaea that have complete or draft genomes is increasing, current genomic information is biased towards species that can be cultivated[96]. Novel techniques, such as single-cell sequencing, are improving access to microbial genomes without isolation[97], but the tree of life remains undersampled for many candidate phyla. In the case of metagenomic analysis, this lack of information hinders the annotation of genes from uncultivated species and introduces uncertainty to comparisons that are made between samples and conditions.

Five metagenomics studies of permafrost have been published so far, although several permafrost metagenomics projects are ongoing from various locations, including Alaska, Greenland and Svalbard, Norway (J.K.J., unpublished observations). The five published studies include a sample from the Canadian High Arctic[46], a core from the Alaskan North Slope[98], two cores from central Alaska before and after 2 days and 7 days of thaw[47], two cores from an upland boreal forest in Alaska plus two cores from the same site after fire (although the upper permafrost layer had thawed after the fire event[20]) and triplicate cores from Sweden[53]. The permafrost metagenomes contained many carbon-cycling and nitrogen-cycling genes from a diverse range of organisms. For example, genes involved in the degradation of carbon compounds, including chitinase, cellobiose, β-glucosidase and β-galactosidase, were detected[20,46], in addition to several genes involved in sugar processing[46,47]; genes encoding trehalose 6-phosphate synthase were twice as abundant in permafrost than in surface soil samples[20]. Trehalose 6-phosphate synthase is an osmoprotectant that catalyses the enzymatic condensation of the precursors glucose-6-phosphate and uridine diphosphate (UDP)-glucose[99]. Trehalose can be used both as a compatible solute and as a carbon source[99]; thus, the presence of genes for trehalose synthesis in permafrost suggests that this could be a life strategy that is used by some permafrost microorganisms to survive under subzero conditions.

Interestingly, functional pathways for both type I and type II methanotrophs and methanogenic archaea have been detected in permafrost metagenomes, which suggests that they have a feedback system for methane production and oxidation[20,47]. However, many of the sequences correspond to uncharacterized members of the community; for example, we previously constructed a draft genome of a novel methanogen from permafrost metagenome sequences[43]. The closest sequenced relative was a member of the Methanocellales (65% sequence identity between genomes), which are known to be metabolically versatile. The draft genome contained the *nifH* gene (which suggests a capacity for nitrogen fixation), the *mcrA* gene, which is required for methanogenesis, and genes for coping with oxidative and temperature stress (specifically, genes that encode catalase, rubredoxin oxidoreductase and the heat-shock protein GroEL). By taking advantage of current high-throughput sequencing capabilities and assembly algorithms, a near-complete genome, similar to the Methanocellales draft genome that we previously identified, was recently constructed from a metagenome that was obtained from a partially thawed permafrost location in Sweden[53]. The resulting Candidatus '*Methanoflorens stordalenmirensis*' spp. genome showed potential for hydrogenotrophic methanogenesis, and this was confirmed by analysing gene expression and carrying out activity measurements in the field[53]. Interestingly, further investigations of existing sequence data from other studies showed that *Methanoflorentaceae*-like Archaea are common in several other high methane-flux habitats, including arctic permafrost regions[53].

Investigation of permafrost metagenomes has also revealed most of the genes encoding proteins that are involved in different steps of the nitrogen cycle, including those involved in nitrate reduction and the assimilation of nitrate and ammonia[20,47]. In the Arctic permafrost metagenomes that have been analysed so far, most of the genes that are involved in the denitrification pathway have been detected, but the relative gene abundances for the last steps in the pathway were too low to lead to N_2 production[20,46]. This could either result in accumulation of N_2O, which is another greenhouse gas, or the assimilation of nitrate into biomass. Moreover, one of the metagenome studies reported large numbers of genes for iron reduction and fermentation, which suggests that these processes may also occur in permafrost[98]. Ultimately, the types of biochemical pathways that are carried out by permafrost microorganisms will be dependent on the available nutrients and redox conditions at any given location, and more metagenome data are needed to be able to make reliable predictions of which processes are likely to occur in specific geographical locations, both now and with climate change.

Metagenomics
Sequencing of total community DNA, thus accessing all genes in the composite genomes of organisms (such as bacteria, archaea, eukarya and viruses) in a given sample, including phylogenetic and protein-coding genes.

Methanotrophs
Aerobic bacteria or anaerobic archaea that are able to metabolize methane (and in some cases other C_1 compounds) as a source of carbon and energy.

Greenhouse gas
A gas in the atmosphere that can absorb and emit radiation within the thermal infrared range. The primary greenhouse gases in the atmosphere of the Earth are carbon dioxide, methane, nitrous oxide, water vapour and ozone.

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Overall, the permafrost metagenome data that are currently available suggest that permafrost is a genetic reservoir that has a large metabolic potential for carbon processing, including pathways for fermentation and methanogenesis. However, there is considerable variation to take into account when analysing replicate permafrost metagenomes from a given location. For example, when data from two adjacent permafrost cores that were collected from either an organic soil[47] or an upland boreal forest[20] were compared, the composition of functional genes was highly variable. These differences included transporter genes for various carbon compounds, which suggests that the cells were exposed to different types of potential substrates[47] and could potentially respond differentially to an environmental perturbation.

Microbial activity in permafrost

Microbial activity has been measured in permafrost in laboratory incubations and *in situ* by measuring gas fluxes or uptake and/or incorporation of labelled substrates[33,50,63,85,100]. Permafrost columns that were incubated with ^{14}C-glucose at temperatures ranging from 0 °C to −33 °C produced two orders of magnitude more $^{14}CO_2$ than autoclaved controls, which is indicative of microbial activity[33]. Steven *et al.*[50] found that ^{14}C-acetic acid and ^{14}C-glucose could be mineralized at −15 °C in a permafrost core that contained ground ice from the Canadian high Arctic. In a recent study, Tuorto *et al.*[101] demonstrated the growth of permafrost bacteria in a subzero temperature range (0 °C to −20 °C). Using stable-isotope probing, the authors detected the incorporation of ^{13}C-acetate into bacterial DNA (from the Acidobacteria, Actinobacteria, Chloroflexi, Gemmatimonadetes and Proteobacteria phyla) during a 6 month-long permafrost incubation experiment. Interestingly, some members of the Acidobacteria, Actinobacteria and Proteobacteria were only able to grow (that is, replicate their genome) when temperatures were lower than −6 °C. Methanotrophic activity was also detected by the measurement of oxidation of ^{14}C-methane to $^{14}CO_2$ at subzero temperatures in ancient permafrost samples (1.8–3 million years old from Siberia)[102]. The maximum rate of oxidation was at the upper permafrost boundary at a depth of 0.4 m. In deeper, more ancient permafrost layers, methane oxidation was between one and two orders of magnitude lower but was still detectable.

By measuring *in situ* CO_2 emissions in the Canadian tundra, it was found that CO_2 fluxes were higher in polygon troughs that are underlain by ice wedges compared with polygon centres[56]. Interestingly, the net production of CO_2 above control levels was observed in late winter, which suggests that continuously active microbial populations were present under subzero conditions. Wilhelm *et al.*[28] suggested that ice-wedge formation is an important determinant of observed microbiological respiration in near subzero temperatures in these environments. This hypothesis was supported by the observation of a build-up of CO_2 in ice wedges and a high rate of flux above a borehole in an ice wedge[103]. Previous studies of *in situ* CO_2 flux report a negative flux, of −3800 mg per m^2 per day from the surface of polygon troughs and −4300 mg per m^2 per day from polygon centres[104], which highlights the range of possible polygon types, some of which have protruding, drier centres, whereas others may have a wetter environment owing to thermokarst development (FIG. 1).

The isotopic signatures of organic depositions can also be used as indicators of microbial activity. The resulting Δ^{13}C- and Δ^{14}C- based CO_2 and CH_4 flux measurements are thus used as an index of *in situ* activity or response to changing conditions[105]. This approach was recently used to document the release of $^{14}CH_4$ to the atmosphere from gas seeps that are concentrated along the boundaries of permafrost thaw and receding glaciers in Alaska and Greenland[106]. The authors mapped 150,000 gas seeps, and a great majority of these released ^{14}C-depleted methane of microbial origin, which was indicative of a reduction in ancient carbon stocks.

Permafrost microbial ecology and climate change

The microbial ecology of permafrost has recently become the focus of intense research efforts owing to concerns about the impact of thaw on the microbial degradation of trapped organic matter, with the potential for the release of the greenhouse gases CO_2, CH_4 and N_2O as a consequence. The total amount of organic carbon that is currently sequestered in permafrost has been estimated to be 1,672 Pg (REF. 107), which is approximately equal to the total amount in vegetation and the atmosphere. As the permafrost thaws and the organic material becomes more accessible to microorganisms, microbial diversity and activity also increase[47,108]. This could result in a large positive feedback to the greenhouse effect.

However, the long-term consequences of permafrost thaw are not well understood[109]. A multiyear permafrost-incubation experiment focusing on modelling CO_2 and CH_4 production from Northeast Siberian Holocene and Pleistocene permafrost deposits showed that, after thaw, a large amount of labile organic matter in the permafrost was readily mineralized[109]. However, model simulations of aerobic and anaerobic CO_2 production over 100 years predicted lower carbon emissions than those that are currently estimated[109]. In addition, owing to low methane-production rates (<0.01 μmol per g dry weight per day), it was difficult to predict a trend that was substantially different from zero rates of emission, and reliable models could not be built. One hypothesis to explain the low predicted rate of carbon emissions over extended time periods was that the carbon in the permafrost deposits was mostly in slowly degrading carbon pools. Further investigation of the quality of carbon deposits in different permafrost sites is necessary to resolve uncertainties concerning the susceptibility of the organic carbon to microbial mineralization.

As mentioned above, several reports have shown that the composition of the microbial community in permafrost differs from that in the seasonally thawed active layer. However, by using a metagenomics sequencing approach, we found that, after only a short-term thaw (of 2 days) at 5 °C, the gene composition shifted rapidly

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Box 3 | **Global efforts to understand the changing Arctic**

Recognition of the potential impact of global warming on permafrost has resulted in the launch of several large research programmes in the United States and Europe.

The US Department of Energy is sponsoring the Next Generation Ecosystem Experiment (NGEE) in the Arctic (NGEE-Arctic Project; see Further information), which is focused on a combination of ecosystem research and modelling to enable better predictions of the impact of climate change in the Arctic. The NGEE-Arctic project is currently carrying out detailed research in Barrow, Alaska — an area that has several permafrost landscape features, including a range of low-centred and high-centred polygons (FIG. 1) and thaw lakes. The microbial ecology questions that are being addressed include: what are the compositions of microbial communities along polygon transects and with depth through the active and permafrost layers? What is the potential for the release of greenhouse gases, such as CO_2, CH_4 and N_2O, as the permafrost thaws owing to microbial activity? Which microbial functions and pathways are actively expressed in frozen permafrost? These questions and others are being addressed using a range of state-of-the-art 'omics' technologies in combination with detailed biogeochemical, geophysical and hydrological data from the site.

Another recently launched programme in Denmark — the Danish Center for Permafrost (CENPERM; see Further information) — is using a similar approach to determine the impact of global warming on permafrost in Greenland.

Similarly, the Norwegian Research Council has recently funded several projects that are focused on the impact of climate change in the global ocean and the melting of the sea ice, with a particular emphasis on the major biogeochemical processes in the Arctic (Climate Change and Its Impacts in Norway (NORKLIMA); see Further Information).

Another type of approach is exemplified by the Earth Microbiome Project (EMP; see Further Information), which is sequencing samples from several regions around the world. Samples from permafrost will be compared with samples that have been collected from other habitats, ranging from host-associated (for example, humans, animals and insects) to marine, soil and plant environments. Ultimately, by comparison of the types of microorganisms in permafrost with other microbial habitats, it should be possible to gain a better understanding of the unique features that enable specific microorganisms to survive and function in permafrost and the features that are responsible for microbial endemism and resilience in the event of climate change.

and started to resemble the composition that is found in the active layer[47]. During thaw, both methane and CO_2 were released. The initial burst of methane was mainly due to the release of trapped gas from the frozen permafrost. Much of the methane was subsequently consumed by methanotrophic bacteria. This example highlights the importance of understanding the sources and sinks of greenhouse gases in a warming climate[108]. The balance between methanogenesis and methane oxidation will ultimately be a determinant of climate feedback. However, much uncertainty remains about how these processes will be affected by thaw in different geographical locations and across different types of permafrost features.

Conclusions and perspectives

The recent International Panel on Climate Change (IPCC) assessment reports[110] forecast a 5–6 °C increase in average surface air temperature in the Arctic by 2099. If correct, this increase in temperature would result in continued changes in Arctic ecosystems, including permafrost thaw. Therefore, it is crucial to understand the probable microbial responses to climate change in high-latitude environments. Although many large-scale efforts are currently in place to develop predictive global climate-change models, these models are generally deficient in taking microbial processes into account. It has already been recognized that insufficient characterization of microbial potential and metabolism in cold environments hampers the power of predictive models[111]. Increasing global efforts have recognized this deficiency and the need to understand the microbial ecology of permafrost (BOX 3). Future research should focus on gaining a better understanding of the types of microorganisms that can survive in different permafrost soils as well as their physiology. By applying high-throughput sequencing techniques and other 'omics' approaches, we now have the ability to identify members of the permafrost microbial community and to understand their functional roles and how they could potentially respond to climate change. However, it is crucial to couple sequencing results with detailed measurements of geochemistry and microbial processes to truly improve our knowledge of these ecosystems. Owing to the complexity of the permafrost soil ecosystem, many variables remain to be assessed, including the impact of permafrost age (or the age of carbon in permafrost) on the microbial carbon-cycling potential. In addition, many indirect effects of microbial responses to their changing environment are yet to be understood. For example, increases in microbial biomass and the availability of new carbon sources are likely to affect interactions at higher trophic levels and food webs, with unknown consequences on ecosystem services.

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Competing interests statement
The authors declare no competing interests.

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